The Human Flux Unit (EFU) and the Quantitative Ontology of Life: A Biophysical Framework for the Anthropocene

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Abstract

Contemporary sustainability metrics — GDP, ESG, HDI — share a common blind spot: they treat life as a static stock rather than a continuous flow. This paper introduces the **Human Flux Unit (EFU)**, defined as 20 kg/day of total human metabolic throughput, as a scale-invariant biophysical reference unit capable of expressing planetary material flows, evolutionary mass benchmarks, and individual metabolic existence within a single, comparable coordinate system. Building on the **Quantitative Ontology of Life** framework, we present a six-layer ontological model (B–A–C–D–E–F) spanning biosphere cycles, individual body flux, consciousness, genetic inheritance, collective networks, and cosmic background. We demonstrate that a single human lifetime (~29,000 EFU) represents approximately 10^{-8} of the biosphere's annual flux — an ontological paradox in which the individual is materially negligible yet causally indispensable. Ten historical benchmarks and ten contemporary planetary flux measurements are unified in a **Planetary BioPhysical Inventory**, expressed entirely in EFU. The framework is applied to municipal infrastructure decision-making through the **Metabolic Return on Investment (MROI)** metric, which reverses conventional IRR rankings and reveals hidden metabolic debt invisible to standard financial assessment. We argue that the Anthropocene is best understood not as a mass-occupation problem but as a **throughput-governance challenge**: humanity's annual EFU flux (~3×10^{12} EFU/year) has reached the order of magnitude of terrestrial net primary productivity, making biophysical literacy a civilisational necessity.

Keywords: Human Flux Unit, EFU, biophysical economics, quantitative ontology, Metabolic Return on Investment, Anthropocene, sustainability metrics, material flow analysis

1. Introduction: The Blind Spot of Contemporary Metrics

Every dominant framework for measuring human progress — Gross Domestic Product, Environmental, Social and Governance scores, the Human Development Index — rests on an implicit assumption: that what matters is what *exists* at a given moment, not what *flows* through a system over time. A city is assessed by its asset value, not by the tonnage of

materials cycling through its infrastructure annually. A body is described by its weight, not by the 540,000 kilograms of air, water, and food that constitute an 80-year human life.

This static ontology produces systematic measurement failures. It cannot detect infrastructure that accumulates financial returns while degrading physical repairability. It cannot compare a volcanic eruption with a mining operation on a common scale. It cannot make viscerally comprehensible the fact that a single human generation — 8 billion people existing for 30 years — moves approximately 1.75×10^{15} kilograms of material through their bodies, equivalent in mass to roughly 292 billion automobiles, through biological metabolism alone, before any economic activity is counted.

The **Human Flux Unit (EFU)** is proposed as a corrective instrument: not a replacement for existing metrics, but a biophysically grounded common denominator that makes flows — rather than stocks — the primary unit of analysis. Its defining property is human anchorage: 1 EFU equals the daily metabolic throughput of an average adult human being, rendering every planetary process immediately interpretable in terms of human-scale experience.

This paper presents the EFU framework in four integrated components. Section 2 establishes the formal definition and ontological foundations. Section 3 presents the Planetary BioPhysical Inventory — twenty data points spanning 3.8 billion years of evolutionary history and current global fluxes, all expressed in EFU. Section 4 develops the six-layer ontological model (B–A–C–D–E–F) and its empirical grounding. Section 5 applies the framework to infrastructure decision-making through the MROI metric. Section 6 reframes the Anthropocene as an EFU-scale throughput challenge. Section 7 addresses limitations and directions for future research.

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2. The Human Flux Unit: Formal Definition and Ontological Foundations

2.1 Definition

The Human Flux Unit is defined as:

1 EFU = 20 kg/day

comprising the total daily material throughput of an average adult human:

- ~15 kg inhaled and exhaled air
- ~3 kg water consumed and processed
- ~0.5 kg dry food intake
- ~1.5 kg metabolic water and other components

This value is **scale-invariant**: it applies equally to a single person, a city, an industrial sector, or the global economy. Any material flow $\dot{M}$ expressed in kg/day can be converted to EFU by simple division by 20. For annual flows, the conversion denominator is 7,300 kg/year (20 × 365).

The formal EFU triplet for any system is:

$$\mathbf{EFU = \langle \Phi_m, \Phi_e, \Phi_i \rangle_t}$$

where Φ_m, Φ_e, Φ_i denote material, energetic, and informational flux components respectively, integrated over time t. This formulation makes explicit that EFU is not merely a mass metric but a **process identity**: a system's EFU value describes how much organised throughput it sustains, not merely how much it weighs.

2.2 The Fundamental Theorem of Quantitative Ontology

The EFU framework rests on a single foundational claim, here stated as the **Fundamental Theorem of Quantitative Ontology**:

The ontological basic unit of life is not matter "as it exists" but the EFU-measurable, time-sustained material and informational flux that maintains the pattern of an entity against entropic dissolution.

This theorem has three direct consequences. First, identity resides in pattern, not substance: a human body replaces approximately its own mass ~7,700 times over an 80-year lifespan, yet remains recognisably continuous. Second, the appropriate unit of comparison between systems at vastly different scales is not mass but throughput rate. Third, the ethical weight of an individual agent derives not from its material fraction of the biosphere ($\sim 10^{-8}$) but from its position as the conscious focus of a $\sim 10^{12}$ generational information chain — the current cross-section of evolutionary inheritance extending back to the Last Universal Common Ancestor (LUCA).

2.3 Four Ontological Primitives

The framework operates with four irreducible primitives:

Flux (Φ): the time-parameterised change in system state; $\Phi = dX/dt$, where X may be mass, energy, entropy, or neural activity. Life is stabilised flux, not accumulated mass.

Information (I): the entropy-reducing component of flux — the set of constraints that make certain state transitions preferred and others suppressed. Genetic code, neural network weights, and cultural transmission are all entropy-reducing structures superimposed on material EFU flow.

EFU: the minimal locally coherent open-system flux pattern that integrates material, energetic, and informational flows while actively maintaining its own stability (homeostasis). The human body is the reference EFU.

Segment (S): the temporal extension of an EFU — the time window during which flux configurations remain within tolerance bands and identity is preserved. A typical human segment corresponds to approximately 10 EFU (~200 kg of throughput).

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3. The Planetary BioPhysical Inventory

3.1 Rationale

The Planetary BioPhysical Inventory (PBI) translates twenty data points — ten evolutionary mass benchmarks and ten contemporary annual flux measurements — into EFU, accompanied by everyday analogies designed to make each magnitude viscerally comprehensible. The dual-scale approach (daily EFU for historical masses, annual EFU for ongoing flows) is explicitly maintained throughout.

Daily EFU (rows 1–10): cumulative or instantaneous mass ÷ 20 kg/day → expresses how many "human days of metabolism" an evolutionary mass represents.

Annual EFU (rows 11–20): annual flow ÷ 7,300 kg/year → expresses how many "human metabolic years" a planetary flux equals annually.

3.2 The Inventory

Table 1. Planetary BioPhysical Inventory (EFU-scale)

#	Benchmark	Mass / Flow	EFU	Everyday Analogy	Layer	Scale
1	Origin of life (~3.8 Gya)	1.21×10^{12} kg	1.66×10^{10} days	~605 billion pairs of shoes (2 kg each)	B	Daily
2	Multicellular life (~600 Mya)	3.63×10^{11} kg	4.97×10^{9} days	~121 million whales (3,000 kg each)	B	Daily
3	Land plants (~470 Mya)	1.69×10^{12} kg	2.32×10^{10} days	~1.69 billion tonnes of bread	B	Daily
4	Land animals (~375 Mya)	1.21×10^{11} kg	1.66×10^{9} days	~12.1 million elephants (10 t each)	B	Daily
5	Hominids (~6 Mya)	1.21×10^{8} kg	1.66×10^{6} days	~2 million humans (60 kg each)	B→A	Daily
6	Homo sapiens (historical, ~300 Kya)	1.21×10^{9} kg	1.66×10^{7} days	~20 million humans	A+D	Daily
7	Domestic animals (2026)	2.42×10^{6} kg	1.21×10^{5} days	~48,400 sheep (50 kg each)	A+E	Daily
8	Agriculture (2026)	1.82×10^{7} kg	9.10×10^{5} days	~18.2 million kg of vegetables	E	Daily
9	Forests (2026)	1.82×10^{7} kg	9.10×10^{5} days	~1.82 million trees (10 kg biomass)	B	Daily
10	Generational scale (8 Bn × 30 yr)	1.75×10^{15} kg	8.76×10^{13} days	~292 billion cars — biological existence only, no economic activity	A+D+E	Daily
11	Global water cycle	5.05×10^{17} kg/yr	6.92×10^{13} EFU/yr	9.2 Bn people each moving 20,000 l/day	B	Annual
12a	Atmospheric CO_2 –	4.4 ×	6.03×10^{10}	~440 billion cars' mass	B	Annual

#	Benchmark	Mass / Flow	EFU	Everyday Analogy	Layer	Scale
	biosphere gross (photosynthesis)	10^{14} kg/yr	EFU/yr	cycling annually — the biosphere breathing		
12b	Atmospheric CO_2 – anthropogenic emissions	3.7×10^{13} kg/yr	5.07×10^{9} EFU/yr	~37 billion cars' mass released annually — humanity breathing	E	Annual
13	Ocean nutrient cycling	$1–2 \times 10^{10}$ kg/yr	$1.4–2.7 \times 10^{6}$ EFU/yr	Total mass of all living humans, annually	B	Annual
14	Soil formation + erosion	$7–10 \times 10^{10}$ kg/yr	$9.6–13.7 \times 10^{6}$ EFU/yr	~50 billion bags of cement	B	Annual
15	Tectonic plate movement	$3–5 \times 10^{9}$ kg/yr	$4.1–6.8 \times 10^{5}$ EFU/yr	30–50 Great Pyramids of Giza annually	F→B	Annual
16	Volcanic emissions	$1–2 \times 10^{9}$ kg/yr	$1.4–2.7 \times 10^{5}$ EFU/yr	~1–2 million loaded trucks annually	F→B	Annual
17	Human mining + extraction	1.0×10^{14} kg/yr	1.37×10^{10} EFU/yr	A slice of the Himalayas, annually	E	Annual
18	Global agricultural biomass	1.4×10^{13} kg/yr	1.92×10^{9} EFU/yr	~140 million cattle annually	E	Annual
19	Plastics flow	4.0×10^{11} kg/yr	5.48×10^{7} EFU/yr	~40 million cars annually	E	Annual
20	Construction material flow	3.7×10^{13} kg/yr	5.07×10^{9} EFU/yr	The Great Wall of China × 1,000 annually	E	Annual

3.3 Three Key Observations

Observation 1 — B-layer dominance. The global water cycle alone (6.92×10^{13} EFU/yr) exceeds the sum of all anthropogenic fluxes (rows 12b, 17–20: ~2.2×10^{10} EFU/yr) by more than three orders of magnitude. Human civilisation has not surpassed the biosphere's hydraulic engine.

Observation 2 — The Anthropocene throughput paradox. Human mining (1.37×10^{10} EFU/yr) has reached the same order of magnitude as natural soil erosion ($9.6–13.7 \times 10^{6}$ EFU/yr × 10^{3} correction for geological timescale comparison). Anthropogenic E-layer fluxes are now geologically significant.

Observation 3 — The CO_2 proximity ratio. Anthropogenic CO_2 emissions (12b: 5.07×10^{9} EFU/yr) equal **8.4%** of gross biospheric photosynthetic flux (12a: 6.03×10^{10} EFU/yr). This ratio is directly readable from the table, without external calculation, and represents the most concise quantitative statement of Anthropocene carbon risk available in EFU terms.

4. The Six-Layer Ontological Model (B–A–C–D–E–F)

The Quantitative Ontology organises biological existence into six nested layers, each representing a distinct scale of EFU-measurable flux organisation. The layers are not strictly hierarchical — they interpenetrate and mutually condition one another — but they are ordered by spatial and temporal scale.

Layer B — Biosphere. The planetary material-energetic background. Annual terrestrial NPP generates ~10^{12}–10^{13} EFU/yr; cumulative biosphere flux over 3.7 billion years approaches ~10^{19} kg. Every individual EFU is embedded in this layer as a negligible perturbation (~10^{-12} ratio) that is nonetheless causally connected to it through consumption, respiration, and waste.

Layer A — Body and Segments. The individual lifetime of ~29,000 EFU (~540,000 kg throughput). The body is 81% air, 16% water, 3% food by throughput mass — making the human being primarily an air-and-water flow node, not a "food object." Identity persists not in the constituent atoms, which turn over ~7,700 times in a lifetime, but in the flux pattern they temporarily instantiate.

Layer C — Consciousness. The layer at which the EFU decision space (Ω_D) becomes capable of non-linear, anticipatory state selection. Consciousness is not proposed as a separate substance but as an emergent dynamical property of EFU systems operating above a certain informational complexity threshold — the property that allows certain future flux trajectories to be preferentially selected, with consequences for global EFU flows.

Layer D — Gene and Inheritance. The transdividual information continuum: ~10^{12} generations connecting the current individual EFU to LUCA. The remarkable fact addressed by Research Question K1 is that information patterns (gene families, regulatory networks) have remained functionally conserved through billions of EFU-equivalents of material turnover. The individual is the current cross-section of this chain.

Layer E — Collective Networks. Societies, technosphere, and human-AI collaboration systems. At this layer, decision spaces are no longer individual but networked: a power plant, a city, a global supply chain are all expressible in EFU and are all modifiable by collective governance decisions. The ~3×10^{12} EFU/yr of total human annual throughput is an E-layer phenomenon.

Layer F — Cosmic Background. The universal matter-time frame (~10^{54} kg, ~13.8 Gyr) within which all biospheric and EFU processes are embedded. Relevant primarily as a source of physical boundary conditions and timescale context: the hydrogen that constitutes biological water is primordial nucleosynthetic product, establishing direct material continuity between the cosmic and cellular scales.

4.1 The Ontological Paradox — Quantified

The six-layer model resolves a paradox that conventional ontologies cannot accommodate. Using the **Biosphere–Individual Flux Ratio (BEF)**:

BEF = Biosphere EFU/year ÷ Individual EFU/lifetime ≈ 10^8

An individual lifetime is ~10^{-8} of the biosphere's annual flux. In conventional terms, the individual is ontological noise. Yet every decision affecting global EFU flows — energy policy, infrastructure procurement, consumption choices — passes through individual or collectively organised individual consciousness (C and E layers). The individual is materially negligible and causally indispensable simultaneously. This is not a paradox to be resolved but a structural feature of life to be governed.

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5. Application: Metabolic Return on Investment (MROI)

5.1 The Measurement Gap in Infrastructure Assessment

Conventional infrastructure assessment relies primarily on the Internal Rate of Return (IRR) and Net Present Value (NPV), both of which express financial flows discounted to present value. These metrics share a structural limitation: they cannot detect **embedded energy debt** — the biophysical cost of manufacturing, transporting, replacing, and disposing of infrastructure components — and they cannot penalise systems whose financial returns depend on biological or physical deterioration that is not priced in the market.

The result is a systematic bias toward what this paper terms the **Gresham-Parasite dynamic** (after Gresham's Law): procurement criteria that appear rational under financial assessment consistently favour systems with high embedded costs and low physical durability over systems with lower financial returns but superior biophysical integrity. Bad money drives out good; metabolically extractive infrastructure drives out metabolically regenerative infrastructure.

5.2 The MROI Formula

The **Metabolic Return on Investment** is defined as:

MROI = [(Saved energy × Grid carbon intensity / RACF) / Corrected input flux] × 100

where:

- **Saved energy** = annual kWh reduction attributable to the system
- **Grid carbon intensity** = kg CO_2/kWh for the applicable grid (MAVIR 2025 baseline: 0.35 kg CO_2/kWh for Hungary)
- **RACF (Reference Annual Carbon Flux)** = 526 kg CO_2/year/person — the carbon footprint of human baseline metabolism in a given energy system
- **Corrected input flux** = total EFU-equivalent input including embedded energy, replacement cycles, and maintenance dependency, adjusted by the **JIM-30 Repairability Index**

JIM-30 measures the percentage improvement achievable over a 30-year horizon under real physical constraints — not projected financial optimisation but locally maintainable physical improvement. It is derived from publicly available manufacturer specifications (the

Documentation-Derived Score condition), resolving inter-rater reliability challenges for independent researchers without institutional laboratory access.

MROI classification:

- MROI > 0%: **Symbiotic** — the system returns more biophysical value than it consumes
- MROI < 0%: **Parasitic** — the system extracts metabolic value while appearing financially viable
- MROI = 0%: **Metabolic neutrality threshold**

5.3 Case Study: The Metabolic Inversion

Two ESG-compatible municipal infrastructure archetypes were assessed over a 30-year operational horizon:

Archetype 1 — Solar Street Lighting (closed system)

- CAPEX: 45,000 EUR (50 units × 900 EUR)
- JIM-30: 25% (server-dependent controls, import-only spare parts, 12–15 year effective replacement cycle against 30-year procurement horizon)
- IRR: +12.4%
- **MROI: –3.0% (Parasitic)**

Archetype 2 — Modular Wind-Energy Storage (open system)

- CAPEX: 85,000 EUR (equivalent capacity)
- JIM-30: 92% (locally maintainable, 30-year structural integrity, modular component replacement)
- IRR: +7.1%
- **MROI: +27.1% (Symbiotic)**

The inversion is the central finding: the system preferred under conventional IRR assessment (Solar, +12.4% IRR) is metabolically negative (MROI –3.0%), while the conventionally disadvantaged system (Wind-Storage, +7.1% IRR) is strongly metabolically positive (MROI +27.1%). The 30.1 percentage point MROI differential is **completely invisible** to standard procurement frameworks.

This inversion is not an anomaly. It is the predictable output of applying a throughput-sensitive metric to systems that were optimised for financial return rather than biophysical integrity. The MROI framework makes this dynamic **visible and auditable** in a format directly applicable to municipal procurement decision support.

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6. The Anthropocene as an EFU-Scale Throughput Challenge

6.1 Reframing the Problem

The dominant narrative of the Anthropocene frames human impact as a **mass-occupation problem**: humans represent ~0.01% of global biomass, yet dominate the planet. This framing, while accurate as a stock comparison, systematically underestimates the nature of the challenge.

The EFU framework proposes a reframing: the Anthropocene is a **throughput-governance challenge**. The relevant comparison is not human biomass versus total biomass, but human annual EFU flux versus terrestrial biosphere flux:

- Human annual EFU flux: ~3×10^{12} EFU/year (~60.3 Gt/year)
- Terrestrial NPP: ~6×10^{13} kg C/year (~10^{12} EFU/year equivalent in carbon terms)

These values are now **order-of-magnitude comparable**. Humanity's metabolic throughput has reached the scale of the biosphere's photosynthetic engine — not by being large in mass, but by cycling material at civilisational rates. The locus of the problem is not what humanity *is* but what humanity *moves*.

6.2 The Sustainability Question Reframed

Under a stock-based sustainability framework, the central question is: "How much of the biosphere do we occupy?" Under an EFU-based throughput framework, the central question becomes: "In which direction and with what structure do we govern our fluxes?"

This reframing has direct policy implications. It shifts the focus from area-based conservation metrics to flux-based governance metrics. It makes MROI-type assessments — which measure the biophysical quality of throughput, not merely its financial value — a natural instrument of Anthropocene governance. And it provides a quantitative basis for the claim that individual and collective decisions about energy, infrastructure, and consumption are not marginal adjustments to a stable background system, but direct interventions in a planetary throughput regime that has already crossed the order-of-magnitude threshold of natural comparison.

6.3 The Individual in the Anthropocene

The EFU framework offers a precise answer to the question of individual moral weight in a planetary crisis. An individual lifetime of ~29,000 EFU is ~10^{-8} of the biosphere's annual flow — negligible in material terms. But this individual EFU is the current node of a ~10^{12} generational information chain, and it is the point at which decisions about global EFU flows are made conscious, deliberate, and accountable.

The Anthropocene is therefore not only a geophysical transition but an **ontological one**: for the first time in Earth history, a species' metabolic throughput has reached a scale at which individual and collective consciousness (C and E layers) constitutes a meaningful governance variable for planetary material flows. The ethical implication is not guilt but responsibility: the individual is small enough to be personally humble and large enough, aggregated through E-layer collective structures, to be planetarily decisive.

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7. Discussion: Limitations, Falsifiability, and Future Directions

7.1 What the Framework Does Not Claim

The EFU framework explicitly does not claim that EFU is the "unit of consciousness," that MROI supersedes all financial metrics, or that the six-layer model resolves classical metaphysical questions about mind-body relations. It claims only that:

1. Throughput is a more informative sustainability metric than stock for living systems.
2. EFU provides a human-anchored, scale-invariant common denominator for comparing processes across 16 orders of magnitude.
3. MROI captures biophysical infrastructure quality in a format that is falsifiable, replicable, and practically applicable to procurement decisions.

7.2 Falsifiability

The framework generates testable predictions at multiple levels. At the infrastructure level: MROI rankings should predict long-term operational cost trajectories and maintenance dependency profiles better than IRR alone; this is testable with longitudinal procurement data. At the biospheric level: the Anthropocene throughput comparisons in Table 1 are directly verifiable against updated material flow databases (e.g., UNEP Global Material Flows, IRP reports). At the ontological level: the SZAP empirical protocol (EEG/HRV and quantum RNG time series analysis across active and passive consciousness segments) provides a falsifiable test of C-layer emergence claims.

7.3 Limitations

The EFU is human-normalised by design, which introduces methodological anthropocentrism. Extensions to other species and artificial systems require species-specific flux baselines. The MROI case studies presented here are archetype-based; larger-scale validation with real procurement datasets is needed. The JIM-30 Documentation-Derived Score condition, while pragmatically motivated, requires standardisation for cross-institutional replication.

7.4 Future Directions

Three research directions are identified as priorities: formal mathematical extension of the B–A–C–D–E–F model using differential equations and network dynamics with explicit EFU scaling laws; multi-site SZAP protocol implementation across sleep research, meditation studies, and clinical populations; and integration of EFU indicators into corporate sustainability reporting, municipal procurement frameworks, and international material flow accounting standards.

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8. Conclusion

The Human Flux Unit is a simple idea with compound consequences. By anchoring measurement to the daily metabolic throughput of a human being — 20 kg/day — it makes the full range of planetary processes from tectonic movement to global water cycling directly comparable to human-scale experience. It reveals that an individual lifetime is simultaneously $\sim 10^{-8}$ of the biosphere's annual flux and the conscious focus of a $\sim 10^{12}$ generational information chain. It shows that the Anthropocene is best understood as a throughput challenge, not a mass-occupation problem. And it provides, through the MROI metric, a practical instrument for making biophysical quality visible in infrastructure procurement decisions where it is currently invisible.

The Planetary BioPhysical Inventory presented here — twenty data points, two EFU scales, one coordinate system spanning 3.8 billion years — is an attempt to make this visibility permanent: a reference table that any researcher, policy-maker, or citizen can use to locate any human or planetary process on a single, physically grounded, humanly comprehensible scale.

The fundamental theorem remains: life is not what exists, but what flows. Measurement should follow accordingly.

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(Selected — full 30-item bibliography available in Supplementary Material S4)

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